

Top 50 AI/ML Interview Questions — Definitions & Key Points

Machine Learning Fundamentals

1. Machine Learning vs AI vs Deep Learning

Machine Learning (ML):

A branch of Artificial Intelligence that enables machines to learn patterns from data and make predictions without explicit programming.

Artificial Intelligence (AI):

A broad field of computer science focused on creating systems that perform tasks requiring human intelligence.

Deep Learning (DL):

A subset of Machine Learning that uses artificial neural networks with multiple layers to learn complex patterns.

Relationship:

AI → Machine Learning → Deep Learning

2. Machine Learning Workflow

Machine Learning Workflow:

A step-by-step process used to build, train, evaluate, and deploy machine learning models.

Steps:

- Data Collection: Gathering required data.
- Data Cleaning: Removing errors and inconsistencies.
- Data Preprocessing: Preparing data for training.
- Feature Engineering: Creating useful features.
- Model Selection: Choosing the right algorithm.
- Model Training: Learning patterns from data.
- Model Evaluation: Measuring performance.
- Model Deployment: Making the model available.
- Model Monitoring: Tracking performance after deployment.

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3. Types of Machine Learning

Supervised Learning

A learning approach where the model learns from labeled data.

Examples:

- Classification
- Regression

Unsupervised Learning

A learning approach where the model finds patterns from unlabeled data.

Examples:

- Clustering
- Dimensionality Reduction

Reinforcement Learning

A learning method where an agent learns through interaction with an environment using rewards and penalties.

Examples:

- Robotics
- Game AI

4. Overfitting and Underfitting

Overfitting

A condition where a model learns training data too closely and performs poorly on unseen data.

Causes:

- Complex model
- Less training data

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- Learning noise

Solutions:

- Regularization
- More data
- Dropout

Underfitting

A condition where a model fails to learn important patterns.

Causes:

- Simple model
- Insufficient training

Solutions:

- Increase model complexity
- Improve features

5. Bias and Variance

Bias

Error caused by incorrect assumptions in the model.

High Bias:

- Simple model
- Underfitting

Variance

Error caused by model sensitivity to training data.

High Variance:

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- Complex model
- Overfitting

Bias-Variance Tradeoff

Balancing model complexity to achieve better generalization.

6. Train-Validation-Test Split

Training Data:

Used for learning model parameters.

Validation Data:

Used for tuning model settings.

Testing Data:

Used for final evaluation.

7. Cross Validation

Cross Validation:

A technique used to evaluate model performance by splitting data into multiple subsets.

K-Fold Cross Validation

Data is divided into K equal parts, and each part is used as validation once.

Stratified K-Fold

Maintains the same class distribution in each fold.

8. Features and Labels

Features:

Input variables used for prediction.

Examples:

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- Age
- Salary
- Location

Labels:

Output variable predicted by the model.

Examples:

- Disease/No Disease
- Price

9. Classification vs Regression

Classification

Predicts categorical values.

Examples:

- Spam detection
- Disease prediction

Regression

Predicts continuous numerical values.

Examples:

- Salary prediction
- Price prediction

10. Model Assumptions

Before training check:

- Data quality
- Missing values

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- Outliers
- Feature correlation
- Data distribution
- Scaling requirements
- Class imbalance
- Data leakage

Data Preprocessing

11. Missing Values

Missing Value Handling:

The process of treating incomplete data.

Methods:

- Remove missing values
- Mean/Median/Mode imputation
- Forward fill
- Backward fill

12. Outliers

Outlier:

A data point significantly different from other observations.

Detection:

- IQR method
- Z-score
- Visualization

Treatment:

- Remove

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- Replace
- Transform

13. Standardization vs Normalization

Standardization:

Scaling data with mean 0 and standard deviation 1.

Normalization:

Scaling values into a fixed range, usually 0 to 1.

14. Feature Engineering

Feature Engineering:

Creating or modifying features to improve model performance.

Benefits:

- Better accuracy
- Improved learning
- Reduced complexity

15. Categorical Encoding

Categorical Encoding:

Converting categorical values into numerical format.

Methods:

- Label Encoding
- One-Hot Encoding
- Target Encoding

16. Data Leakage

Data Leakage:

When information from outside training data affects model training.

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Effect:

- Unrealistic accuracy
- Poor real-world performance

17. Imbalanced Dataset Handling

Imbalanced Dataset:

Dataset where one class contains significantly more samples.

Methods:

- Oversampling
- Undersampling
- SMOTE
- Class weights

18. Feature Selection

Feature Selection:

Selecting important features for model training.

Methods:

- Filter Methods
- Wrapper Methods
- Embedded Methods

Machine Learning Algorithms

19. Linear Regression

A supervised algorithm used to predict continuous values by finding linear relationships between variables.

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20. Linear Regression vs Logistic Regression

Linear Regression:

- Used for numerical prediction.

Logistic Regression:

- Used for classification.
- Predicts probability.

21. Decision Tree

A tree-based algorithm that makes decisions using feature-based splits.

Entropy:

Measures uncertainty in data.

Gini Index:

Measures impurity in decision nodes.

22. Random Forest

An ensemble algorithm that combines multiple decision trees to improve accuracy.

23. XGBoost

A gradient boosting algorithm that builds models sequentially to correct previous errors.

24. Support Vector Machine

A supervised algorithm that finds the optimal boundary between classes.

25. K-Nearest Neighbors

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A supervised algorithm that predicts based on similarity with nearby data points.

26. Naive Bayes

A probabilistic classification algorithm based on Bayes theorem.

27. K-Means Clustering

An unsupervised algorithm that groups similar data points into clusters.

28. Bagging vs Boosting

Bagging:

Multiple models trained independently.

Example:

- Random Forest

Boosting:

Models trained sequentially to correct errors.

Examples:

- XGBoost
- AdaBoost

29. PCA

Principal Component Analysis:

A dimensionality reduction technique that converts high-dimensional data into fewer important components.

30. Algorithm Selection

Depends on:

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- Problem type
- Dataset size
- Complexity
- Accuracy requirements
- Interpretability

Model Evaluation

31. Confusion Matrix

A table showing actual and predicted classification results.

Components:

- True Positive
- True Negative
- False Positive
- False Negative

32. Evaluation Metrics

Accuracy:

Percentage of correct predictions.

Precision:

Correctness of positive predictions.

Recall:

Ability to identify actual positives.

F1 Score:

Balance between precision and recall.

33. Accuracy Limitations

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Accuracy is not suitable when:

- Dataset is imbalanced.
- False negatives are costly.
- Rare events are important.

34. ROC and AUC

ROC Curve:

Shows classifier performance at different thresholds.

AUC:

Measures area under ROC curve.

35. Regression Metrics

MAE:

Average absolute prediction error.

MSE:

Average squared prediction error.

RMSE:

Square root of MSE.

36. Hyperparameter Tuning

Process of finding the best model settings.

Methods:

- Grid Search
- Random Search
- Bayesian Optimization

Deep Learning

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37. Artificial Neural Network

A computing model inspired by the human brain.

Layers:

- Input Layer
- Hidden Layer
- Output Layer

38. Backpropagation

Algorithm used to update neural network weights by reducing prediction error.

Steps:

- Forward propagation
- Loss calculation
- Backward propagation
- Weight update

39. CNN vs RNN vs Transformer

CNN:

- Used for images.
- Performs feature extraction.

RNN:

- Used for sequential data.

Transformer:

- Uses attention mechanism for language tasks.

40. Why CNN Works for Images

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Reasons:

- Local connectivity
- Parameter sharing
- Feature extraction
- Translation invariance

41. Vanishing Gradient Problem

A problem where gradients become too small during training.

Solutions:

- ReLU activation
- Batch normalization
- ResNet

42. Dropout

A regularization technique that randomly disables neurons during training to prevent overfitting.

NLP and LLMs

43. Transformer Architecture

A deep learning architecture based on attention mechanisms.

Components:

- Encoder
- Decoder
- Self-Attention
- Positional Encoding

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44. Self-Attention

A mechanism that helps models understand relationships between words.

Components:

- Query
- Key
- Value

45. RAG

Retrieval-Augmented Generation:

A technique that combines retrieval systems with LLMs to generate accurate answers.

Pipeline:

- Document Collection
- Chunking
- Embedding
- Vector Database
- Retrieval
- Generation

46. LLM vs Fine-Tuning vs Prompt Engineering vs RAG

LLM:

Large AI model trained on massive datasets.

Prompt Engineering:

Creating better instructions for AI models.

Fine-Tuning:

Training an existing model on specialized data.

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RAG:

Providing external knowledge during generation.

Deployment and MLOps

47. Model Deployment

The process of making a trained ML model available for real-world predictions.

Steps:

- Save model
- Create API
- Deploy
- Monitor

48. Model Drift

Reduction in model performance due to changes in data or patterns.

Types:

Data Drift

Input data distribution changes.

Concept Drift

Relationship between input and output changes.

49. AI/ML Tools

Common tools:

- Python
- NumPy
- Pandas

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- Scikit-learn
- TensorFlow
- Keras
- PyTorch
- Flask
- FastAPI
- Docker
- Git/GitHub
- Cloud Platforms

50. AI/ML Project Explanation

A structured explanation of a machine learning project.

Points:

- Problem Statement
- Dataset
- Technologies Used
- Methodology
- Model
- Evaluation
- Deployment
- Challenges
- Future Improvements